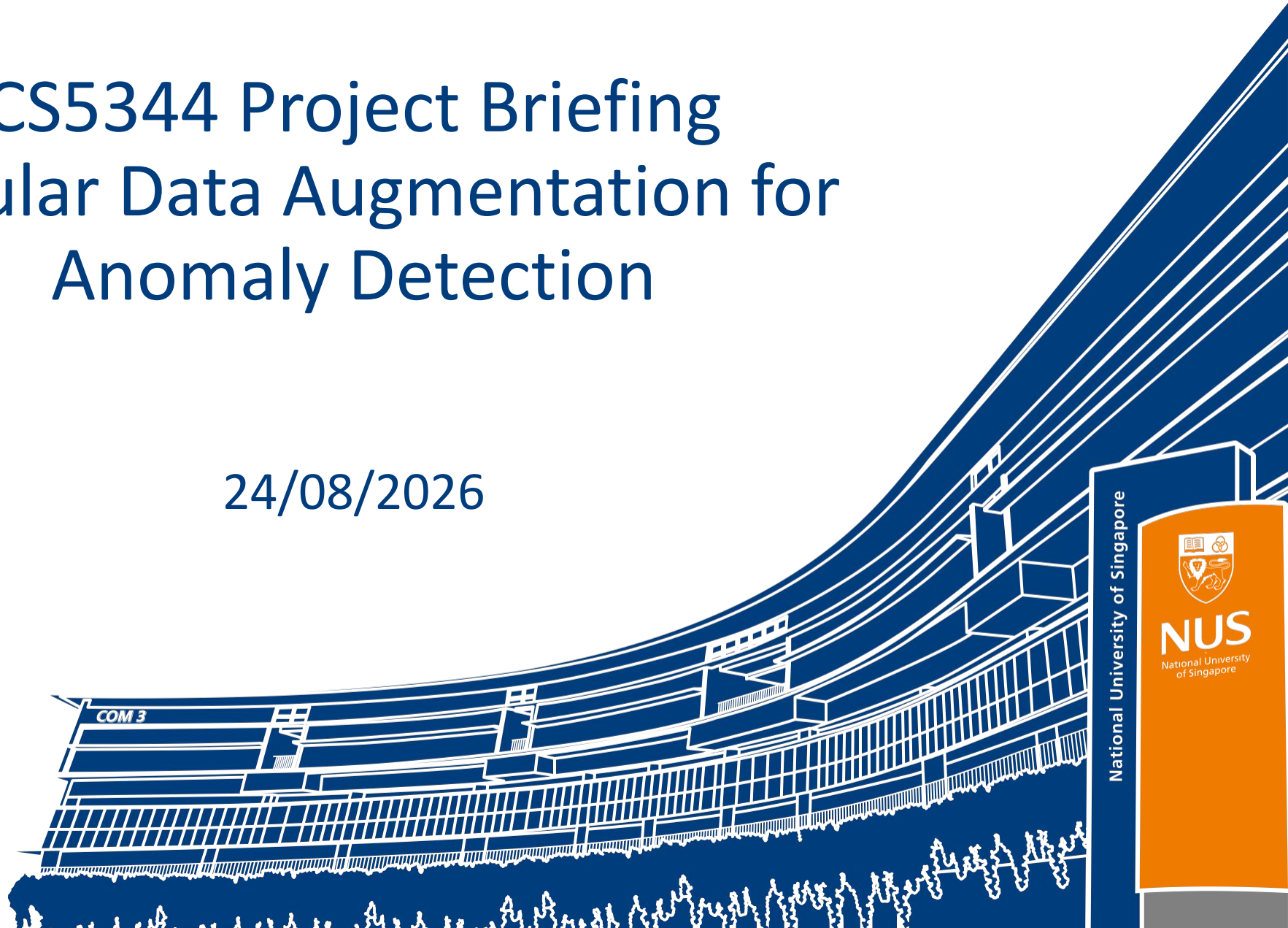


CS5344 Project Briefing

Tabular Data Augmentation for Anomaly Detection

24/08/2026



Outline

- Background & Project Overview
- Problem Formulation
- Dataset Peek
- Kaggle Leaderboard & Grading
- Starter Kit Walkthrough
- Q&A

Background & Project Overview

What is Anomaly Detection?

- Finding data points that deviate from the expected “normal” pattern.
- Widely applied in fraud detection, industrial monitoring, network attack detection...



Fraud Detection

Credit-card / bank transaction records
Flag transactions that don't match a customer's usual spending pattern



Industrial Monitoring

Sensor readings from factory machines
Catch early signs of equipment failure before breakdown



Network Attack Detection

Network connection / traffic logs
Identify intrusions and attacks among normal traffic

Background & Project Overview

Tabular data may contain multiple different types of anomalies, such as:

- Odd Value: a single value that moves far from its usual range.
- Rare Combination: the reading form a combination the record seldom shows, even if each individual value looks normal.
- ...

ID	Age	Annual Income	Transaction Amt	Label
1	34	62,000	120	Normal
2	41	85,000	300	Normal
3	27	45,000	90	Normal
4	500	58,000	150	Abnormal
5	19	59,000	200	Abnormal

Odd Value

Rare
combination

Background & Project Overview

Why augmentation for anomaly detection?



Sharper Boundary

Well-designed synthetic normal data can help the detector learn a clearer normal/abnormal boundary.



Privacy Protection

Real data (e.g. attack logs) is often too sensitive to release directly.

Synthetic data lets us share useful data without exposing real records.



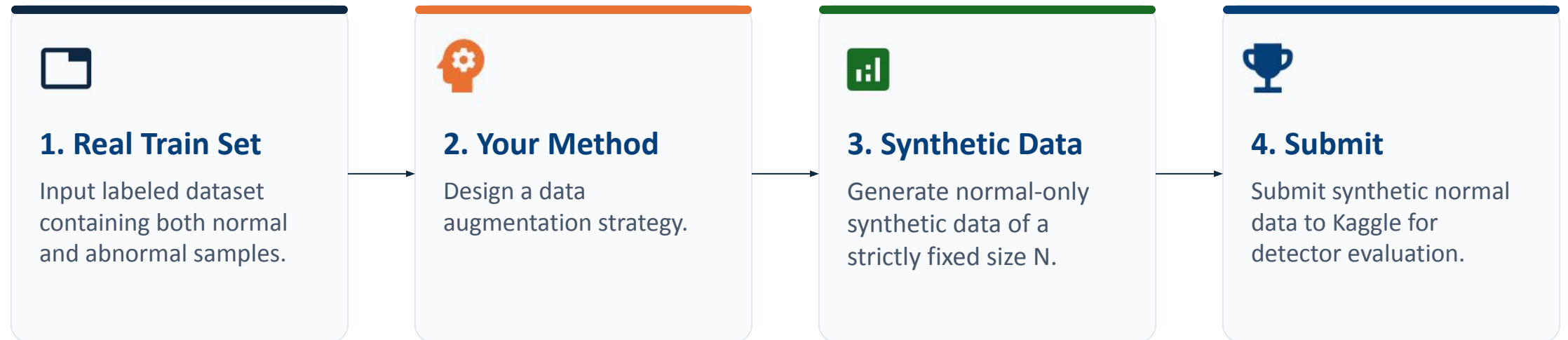
Data Scarcity

Real (especially normal) data may be limited in quantity.

Augmentation fills out and completes the underlying distribution.

Problem Formulation

What will you do in this project?



Problem Formulation

Labeled dataset

- $D = \{x_i\}, i = 1..n$ — n samples, m feature columns (numerical or categorical) per sample
- Each sample x_i has a binary anomaly label $y_i \in \{0, 1\}$: 0 = normal, 1 = abnormal

Train / Validation / Test split

- $(D_{\text{train}}, y_{\text{train}}), (D_{\text{valid}}, y_{\text{valid}}), (D_{\text{test}}, y_{\text{test}})$ — disjoint row partition
- The train set is fully labeled, including both normal and abnormal samples, and only this train set is released to participants.
- Validation / test sets are private, contain both normal and abnormal samples — **including anomaly types never seen during training.**

Problem Formulation

Augmentation Process

- Design a augmentation method using the released ($D_{\text{train}}, y_{\text{train}}$)
- Produce a synthetic dataset $\tilde{D} = \{\tilde{x}_i\}, i = 1..N$, where N is a fixed submission size
- **The submission must consist EXCLUSIVELY of samples meant to represent the normal distribution — no anomaly labels are submitted.**

Performance Evaluation

- A fixed set of anomaly detectors is trained on your submitted $\tilde{D}$, under a fixed training scheme set by the organizers.
- Trained detectors score the private validation / test sets.
- Metrics: AUPRC
- **Our goal is downstream utility, NOT distribution fidelity.**

NSL-KDD

- A refined version of the KDD Cup 1999 network intrusion dataset.
- Each row = one network connection. Features describe connection properties (duration, byte counts, protocol, ...). Attacks span many types, e.g. neptune (SYN-flood DoS), ipsweep (probing/surveillance).

duration	protocol_type	service	src_bytes	dst_bytes	count	...	is_anomaly
0	udp	domain_u	44	134	71	...	0
0	tcp	smtp	2071	332	1	...	0
0	udp	domain_u	45	131	196	...	0
0	tcp	smtp	1505	329	1	...	0
0	icmp	eco_i	18	0	1	...	1
0	tcp	exec	0	0	131	...	1
...	...	...	...	...	...	...	...

m = 41 features (38 numerical + 3 categorical). is_anomaly = 1 rows (orange) are attacks.

UNSW-NB15

- Generated with the IXIA PerfectStorm tool at the Cyber Range Lab of UNSW Canberra.
- Each row = one network flow. Features describe flow properties (packet counts, TCP window size, connection state, ...). Attacks span several categories, e.g. Generic (block-cipher attack), DoS (denial-of-service).

proto	service	state	sttl	swin	ct_srv_dst	...	is_anomaly
udp	dns	CON	31	0	2	...	0
tcp	-	FIN	31	255	10	...	0
tcp	ssh	FIN	31	255	1	...	0
tcp	http	FIN	62	255	2	...	0
tcp	-	FIN	254	255	2	...	1
tcp	http	FIN	62	255	1	...	1
...	...	...	...	...	...	...	...

m = 42 features (39 numerical + 3 categorical); “-” in service = not a well-known service.

Kaggle Leaderboard & Grading

How your synthetic data becomes a leaderboard score



Fixed detectors

ECOD: deterministic tail/rank detector
IForest: 10 fixed seeds × 500 trees
Same settings for every team

Primary Kaggle score

$$S = (\text{AUPRC_ECOD} + \text{AUPRC_IForest}) / 2$$

Important

Submit normal-only synthetic feature rows
Score measures downstream utility

Goal: generate synthetic normal data that helps fixed anomaly detectors separate normal from abnormal hidden samples.

Kaggle Leaderboard & Grading

Public leaderboard gives feedback, and private leaderboard determines the final ranking

Same submitted synthetic data + same fixed detectors

During the competition: Public

Hidden validation rows
Score shown immediately
Used for development feedback

After the competition: Private

Hidden final test rows
Score hidden during competition
Used for final leaderboard ranking

Do not overfit to the public leaderboard. Hidden test rows may include unseen anomaly types.

Kaggle Leaderboard & Grading

Leaderboard performance is evidence, but not a direct grade conversion

Project component (60%)

Proposal writeup: 15%
Final presentation: 20%
Final report: 25%

**Kaggle rank is NOT
converted directly
into marks**

What we grade overall

Leaderboard performance
Method novelty and transferability
Explainability and robustness
Experiments and result analysis
Writing and presentation quality
Reproducibility of submitted code

Rules to remember

Use one transferable pipeline for both datasets (parameter settings can be different).
Deep learning methods are allowed, but you must clearly explain the distributional properties of the generated data and how your method produces them.
Submit source code after competition ends for reproducibility.

Starter Kit Walkthrough

Q&A

- Key things to note
 - Starter kit and problem formulation will be available soon this week.
 - You'll form groups (Week 4 finalization).
 - Kaggle competition opens Week 4 with complete dataset (Limited submission number every day).
 - Between now and then, familiarize yourself with the tasks and settings.
- TAs
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